Iman Gholami

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LinkedIn | GitHub | Scholar

Industry Experience

Intrinsic Technologies (Axiom)
Software Engineer Intern — Jun – Aug 2026, New York, NY

- Developed machine-checked proofs in Lean 4 for Zero-Knowledge Virtual Machines (ZK VMs), strengthening correctness guarantees.
- Designed and implemented an RL evaluation system to assess LLMs and reasoning models on automated proof-generation tasks and guide model optimization.

Axal (YC W25)
Software Engineer Intern — May – Aug 2025, San Francisco, CA

- Architected and built a platform combining static analysis, code dependency graphs, and LLM reasoning from the ground up to automate modernization of large-scale legacy systems.

Balad (30M+ installs)
Software Engineer — Nov 2018 – Dec 2022, Tehran, Iran

- Architected a real-time traffic prediction system, improving ETA accuracy by 35%.
- Overhauled analytics (ClickHouse, Airflow, Spark), cutting compute costs by 30%.
- Migrated the core navigation service from Java to Kotlin, cutting latency by 30%.

Bazaar (100M+ installs)
Software Engineer Intern — Jul – Oct 2018, Tehran, Iran

- Built the internal wallet for 50M+ users; resolved a production issue that had frozen $20,000 in user funds.

Research Experience

University of Maryland
Research Assistant — Jan 2023 – Present, College Park, MD

- Research on network design, including approximation algorithms for Steiner Forest and related connectivity problems.
- Research on formal verification in Lean 4 and LLM-driven proof generation for challenging mathematical problems through **DARPA mathExp**.
- Research on quantum-inspired solvers for hard SAT instances through **DARPA QuICC**.

Education

University of Maryland
PhD in Computer Science
Expected May 2028

Sharif University of Technology
BSc in Computer Engineering
May 2022

Technical Skills

Languages: Python, C++, Lean 4, SQL, TypeScript, Go, Java, Kotlin

AI & Evaluation: LLM agents, model/RL evaluation, automated theorem proving

Research: Network design, approximation algorithms, formal verification, static analysis

Systems: Distributed systems, software architecture, performance optimization

Tools: Docker, Kubernetes, AWS, PostgreSQL, MongoDB, Redis, Git, CI/CD, Linux; FastAPI, Django, React

Selected Open Source

Metachamber
AI-powered metadata discovery and automated data catalog generation.

Grass
Reduced TA grading effort by 90% with rubric-based scoring and LMS integration.

Honors & Awards

FOCS 2025 | Best Paper Award

IOI 2017 | Silver Medal

INOI 2016 | Gold Medal

ICPC Regional | 2nd Place
2017, 2023

Selected Publications

The Tight Competitive Ratio of Greedy for Online Steiner Forest | Lean proofs — **arXiv, 2026**
I. Gholami, I. Hajiaghayi, M. Hajiaghayi, P. Jabbarzade.$^{\alpha\beta}$

Beyond the Library: An Agentic Framework for Autoformalizing Research Mathematics — **arXiv, 2026**
A. Soltani Moakhar, **I. Gholami**, M. Springer, M. JafariRaviz, M. Hajiaghayi.

Bi-Criteria Metric Distortion — **ICLR 2026**
K. Banihashem, D. Chakraborty, S. Chashm Jahan, **I. Gholami**, M. Hajiaghayi, M. Mahdavi, M. Springer.$^{\alpha\beta}$

Breaking a Long-Standing Barrier: 2-ε Approximation for Steiner Forest (Best Paper Award) — **FOCS 2025**
A. Ahmadi, **I. Gholami**, M. Hajiaghayi, P. Jabbarzade, M. Mahdavi.$^{\alpha\beta}$

Prize-Collecting Forest with Submodular Penalties: Improved Approximation — **IPCO 2025**
A. Ahmadi, **I. Gholami**, M. Hajiaghayi, P. Jabbarzade, M. Mahdavi.$^{\alpha\beta}$

2-Approximation for Prize-Collecting Steiner Forest — **JACM 2025 | SODA 2024**
A. Ahmadi, **I. Gholami**, M. Hajiaghayi, P. Jabbarzade, M. Mahdavi.$^{\alpha\beta}$

Prize-Collecting Steiner Tree: A 1.79 Approximation — **STOC 2024**
A. Ahmadi, **I. Gholami**, M. Hajiaghayi, P. Jabbarzade, M. Mahdavi.$^{\alpha\beta}$

$\alpha\beta$ = alphabetical author order.

Full list: Google Scholar